BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 075 · Lesson 075 · Big O: Quadratic, Exponential and Growth

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does O(n) differ from O(1)?

Q2. How many halvings 32→1?

Q3. Which algorithm naturally explores include/exclude branches?

Q4. A rectangular 4×3 zero-indexed array holds how many elements? Give its last valid index pair.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

quadratic — Growth proportional to n^2.

exponential — Growth like 2^n.

pruning — Avoid exploring impossible branches.

scalability — Handling larger inputs feasibly.

New learning

Observe — Two nested n loops perform n×n operations, producing $O(n^2)$ time when each inner operation is constant.

Explain — A binary decision at each of n positions can create 2^n possibilities. This exponential growth quickly outruns polynomial methods.

Evaluate — Big O ignores constants and lower-order terms; compare classes by input growth while acknowledging memory, workload and implementation detail.

Worked example

For n=6, calculate total iterations of two independent nested n loops.

1. Explain — Each of the six outer iterations runs six inner iterations.

We Do · Practise with guidance

For n=6 decisions, calculate binary include/exclude combinations.

`$2^6$=64 possibilities, O($2^n$) in full enumeration.`

Explain

Every new independent binary choice doubles the count.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

For n=5 and n=10, calculate n^2 and 2^n values. [6]

Question 2

Explain time growth of a nested n×n loop and classify. [6]

Question 3

A problem has 30 binary choices. Evaluate full subset enumeration and one way to reduce exploration. [6]

Exit Ticket · Answer before checking

An $O(2^n)$ approach grows only twice as much when n doubles.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. One grows with n; the other does not.

Q2. Five.

Q3. Subset backtracking or exhaustive enumeration.

Q4. 12 elements; last index (3,2).

You Do pages · Independent questions

Q1. n5 $n^2$25 [1] $2^5$32 [1]; n10 $n^2$100 [2] 2^{10}1024 [2].

Q2. Outer n iterations [2], inner n per outer [2], n^2 total [1], O(n^2) [1].

Q3. 2^{30}≈1,073,741,824 subsets [2]; impractical exhaustive work may occur [1]; backtracking pruning with valid bound [2], depends on safe assumptions [1].

Exit Ticket · Model answer

False. It can grow dramatically faster.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: